10 The parametric equations of a curve are

$x = 2 + 5\cos\theta$ and $y = 1 + 5\sin\theta$, where $0 \leqslant \theta \leqslant 2\pi$.

(a) Determine the cartesian equation of the curve. **[3]**

(b) Hence or otherwise, find the equation of the tangent to the curve at the point $(5, -3)$, giving your answer in the form $ax + by + c = 0$, where a, b and c are integers to be determined. **[4]**

10(a)

10(b)